

MSSQL

```
/* =====  
MSSQL LAB: Telecom Data Mapping  
FULL UPDATED VERSION WITHOUT GO  
Execute section by section in SSMS  
===== */
```

```
-----  
-- 1. CREATE DATABASE  
-----
```

```
DROP DATABASE IF EXISTS TelecomDataMappingLab;
```

```
CREATE DATABASE TelecomDataMappingLab;
```

```
USE TelecomDataMappingLab;
```

```
SELECT DB_NAME();  
-----
```

```
-- 2. CREATE TABLES  
-----
```

```
-- Customers Table
```

```
CREATE TABLE customers (  
    customer_id INT IDENTITY(1,1) PRIMARY KEY,  
    customer_full_name VARCHAR(100) NOT NULL,  
    nrc_number VARCHAR(50),  
    gender VARCHAR(10),  
    city VARCHAR(50),  
    created_at DATETIME DEFAULT GETDATE()  
);
```

```
-- Subscribers Table
```

```
CREATE TABLE subscribers (  
    subscriber_id INT IDENTITY(1,1) PRIMARY KEY,  
    customer_id INT NOT NULL,  
    msisdn VARCHAR(20) NOT NULL UNIQUE,  
    sim_number VARCHAR(30) NOT NULL,  
    package_type VARCHAR(50),  
    activation_date DATE,  
    status VARCHAR(20),
```

```
    CONSTRAINT fk_subscriber_customer  
    FOREIGN KEY (customer_id)  
    REFERENCES customers(customer_id)  
);
```

```
-- Recharge Transactions Table
```

```
CREATE TABLE recharge_transactions (  
    recharge_id INT IDENTITY(1,1) PRIMARY KEY,  
    subscriber_id INT NOT NULL,  
    recharge_amount DECIMAL(12,2) NOT NULL,  
    recharge_channel VARCHAR(50),  
    recharge_created_at DATETIME DEFAULT GETDATE(),
```

```
    CONSTRAINT fk_recharge_subscriber  
    FOREIGN KEY (subscriber_id)  
    REFERENCES subscribers(subscriber_id)  
);
```

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-- Data Usage Table

```
CREATE TABLE data_usage (
    usage_id INT IDENTITY(1,1) PRIMARY KEY,
    subscriber_id INT NOT NULL,
    usage_date DATE NOT NULL,
    total_data_mb DECIMAL(12,2) NOT NULL,
    network_type VARCHAR(20),

    CONSTRAINT fk_usage_subscriber
    FOREIGN KEY (subscriber_id)
    REFERENCES subscribers(subscriber_id)
);
```

-- Data Mapping Catalog Table

```
CREATE TABLE data_mapping_catalog (
    mapping_id INT IDENTITY(1,1) PRIMARY KEY,
    business_term VARCHAR(100) NOT NULL,
    source_system VARCHAR(100),
    database_table VARCHAR(100),
    database_field VARCHAR(100),
    field_description VARCHAR(255),
    data_type VARCHAR(50),
    data_owner VARCHAR(100)
);
```

 -- 3. INSERT SAMPLE DATA

-- Customers

```
INSERT INTO customers
(customer_full_name, nrc_number, gender, city)
VALUES
('Aung Aung', '12/ABC(N)123456', 'Male', 'Yangon'),
('Su Su', '10/XYZ(N)234567', 'Female', 'Mandalay'),
('Kyaw Kyaw', '8/MNO(N)345678', 'Male', 'Naypyitaw'),
('Hla Hla', '7/PQR(N)456789', 'Female', 'Bago'),
('Mya Mya', '9/STU(N)567890', 'Female', 'Yangon');
```

-- Subscribers

```
INSERT INTO subscribers
(customer_id, msisdn, sim_number, package_type, activation_date, status)
VALUES
(1, '09790000001', 'SIM100001', 'Prepaid', '2025-01-10', 'Active'),
(2, '09790000002', 'SIM100002', 'Postpaid', '2025-02-15', 'Active'),
(3, '09790000003', 'SIM100003', 'Prepaid', '2025-03-20', 'Inactive'),
(4, '09790000004', 'SIM100004', 'Prepaid', '2025-04-05', 'Active'),
(5, '09790000005', 'SIM100005', 'Postpaid', '2025-05-12', 'Active');
```

-- Recharge Transactions

```
INSERT INTO recharge_transactions
(subscriber_id, recharge_amount, recharge_channel, recharge_created_at)
VALUES
(1, 5000, 'Mobile Wallet', '2026-05-01 09:30:00'),
(1, 7000, 'Retail Shop', '2026-05-03 14:20:00'),
(2, 15000, 'Bank App', '2026-05-02 10:10:00'),
(3, 3000, 'Retail Shop', '2026-05-04 16:45:00'),
(4, 12000, 'Mobile Wallet', '2026-05-05 11:00:00'),
(5, 20000, 'Bank App', '2026-05-06 18:30:00');
```

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(1, 10000, 'Mobile Wallet', '2026-05-07 08:00:00');

-- Data Usage

INSERT INTO data_usage

(subscriber_id, usage_date, total_data_mb, network_type)

VALUES

(1, '2026-05-01', 1200.50, '4G'),

(1, '2026-05-02', 850.25, '4G'),

(2, '2026-05-02', 2300.75, '5G'),

(3, '2026-05-03', 400.00, '3G'),

(4, '2026-05-04', 1750.30, '4G'),

(5, '2026-05-05', 3200.90, '5G'),

(1, '2026-05-06', 1500.00, '4G');

-- Data Mapping Catalog

INSERT INTO data_mapping_catalog

(business_term, source_system, database_table, database_field, field_description, data_type, data_owner)

VALUES

('Customer Name', 'CRM System', 'customers', 'customer_full_name', 'Full name of telecom customer', 'VARCHAR(100)', 'CRM Team'),

('Customer NRC', 'CRM System', 'customers', 'nrc_number', 'National registration number', 'VARCHAR(50)', 'CRM Team'),

('Customer City', 'CRM System', 'customers', 'city', 'Customer registered city', 'VARCHAR(50)', 'CRM Team'),

('Mobile Number', 'Billing System', 'subscribers', 'msisdn', 'Customer mobile phone number', 'VARCHAR(20)', 'Billing Team'),

('SIM Number', 'Billing System', 'subscribers', 'sim_number', 'SIM card serial number', 'VARCHAR(30)', 'Billing Team'),

('Package Type', 'Billing System', 'subscribers', 'package_type', 'Prepaid or postpaid package type', 'VARCHAR(50)', 'Product Team'),

('Recharge Amount', 'Billing System', 'recharge_transactions', 'recharge_amount', 'Top-up amount by customer', 'DECIMAL(12,2)', 'Billing Team'),

('Recharge Date', 'Billing System', 'recharge_transactions', 'recharge_created_at', 'Date and time of recharge', 'DATETIME', 'Billing Team'),

('Recharge Channel', 'Billing System', 'recharge_transactions', 'recharge_channel', 'Channel used for recharge', 'VARCHAR(50)', 'Sales Team'),

('Internet Usage', 'Network System', 'data_usage', 'total_data_mb', 'Total internet usage in MB', 'DECIMAL(12,2)', 'Network Team'),

('Usage Date', 'Network System', 'data_usage', 'usage_date', 'Date of internet usage', 'DATE', 'Network Team'),

('Network Type', 'Network System', 'data_usage', 'network_type', '3G, 4G, or 5G network type', 'VARCHAR(20)', 'Network Team');

156 SELECT * FROM customers;

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customer_id	customer_full_name	nrc_number	gender	city	created_at
1	Aung Aung	12ABC(N)123456	Male	Yangon	2026-05-10 22:18:06.217
2	Su Su	10XYZ(N)234567	Female	Mandalay	2026-05-10 22:18:06.217
3	Kyaw Kyaw	8MNO(N)345678	Male	Naypyitaw	2026-05-10 22:18:06.217
4	Hla Hla	7PQR(N)456789	Female	Bago	2026-05-10 22:18:06.217
5	Mya Mya	9STU(N)567890	Female	Yangon	2026-05-10 22:18:06.217

157 SELECT * FROM subscribers;

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

subscriber_id	customer_id	msisdn	sim_number	package_type	activation_date	status
1	1	09790000001	SIM100001	Prepaid	2025-01-10	Active
2	2	09790000002	SIM100002	Postpaid	2025-02-15	Active
3	3	09790000003	SIM100003	Prepaid	2025-03-20	Inactive
4	4	09790000004	SIM100004	Prepaid	2025-04-05	Active
5	5	09790000005	SIM100005	Postpaid	2025-05-12	Active

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160 SELECT * FROM recharge_transactions;

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

recharge_id	subscriber_id	recharge_amount	recharge_channel	recharge_created_at
1	1	5000.00	Mobile Wallet	2026-05-01 09:30:00.000
2	1	7000.00	Retail Shop	2026-05-03 14:20:00.000
3	2	15000.00	Bank App	2026-05-02 10:10:00.000
4	3	3000.00	Retail Shop	2026-05-04 16:45:00.000
5	4	12000.00	Mobile Wallet	2026-05-05 11:00:00.000
6	5	20000.00	Bank App	2026-05-06 18:30:00.000
7	1	10000.00	Mobile Wallet	2026-05-07 08:00:00.000

161 SELECT * FROM data_usage;

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

usage_id	subscriber_id	usage_date	total_data_mb	network_type
1	1	2026-05-01	1200.50	4G
2	1	2026-05-02	850.25	4G
3	2	2026-05-02	2300.75	5G
4	3	2026-05-03	400.00	3G
5	4	2026-05-04	1750.30	4G
6	5	2026-05-05	3200.90	5G
7	1	2026-05-06	1500.00	4G

164 SELECT * FROM data_mapping_catalog;

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

mapping_id	business_term	source_system	database_table	database_field	field_description	data_type	data_owner
4	Mobile Number	Billing System	subscribers	msisdn	Customer mobile phone num...	VARCHAR(20)	Billing Te...
5	SIM Number	Billing System	subscribers	sim_number	SIM card serial number	VARCHAR(30)	Billing Te...
6	Package Type	Billing System	subscribers	package_type	Prepaid or postpaid package ...	VARCHAR(50)	Product T...
7	Recharge Amount	Billing System	recharge_tran...	recharge_amount	Top-up amount by customer	DECIMAL(12,2)	Billing Te...
8	Recharge Date	Billing System	recharge_tran...	recharge_created_at	Date and time of recharge	DATETIME	Billing Te...
9	Recharge Channel	Billing System	recharge_tran...	recharge_channel	Channel used for recharge	VARCHAR(50)	Sales Team
10	Internet Usage	Network Syst...	data_usage	total_data_mb	Total internet usage in MB	DECIMAL(12,2)	Network T...
11	Usage Date	Network Syst...	data_usage	usage_date	Date of internet usage	DATE	Network T...

166 -- 5. CUSTOMER + SIM INFORMATION

167

168

169

170 SELECT

171 c.customer_full_name AS Customer_Name,

172 c.city,

173 s.msisdn AS Mobile_Number,

174 s.sim_number,

175 s.package_type,

176 s.status

177 FROM customers c

178 INNER JOIN subscribers s

179 ON c.customer_id = s.customer_id;

180

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

customer_full_name	msisdn	recharge_amount	recharge_channel	recharge_created_at
1 Aung Aung	09790000001	5000.00	Mobile Wallet	2026-05-01 09:30:00.000
2 Su Su	09790000002	15000.00	Bank App	2026-05-02 10:10:00.000
3 Aung Aung	09790000001	7000.00	Retail Shop	2026-05-03 14:20:00.000
4 Kyaw Kyaw	09790000003	3000.00	Retail Shop	2026-05-04 16:45:00.000
5 Hla Hla	09790000004	12000.00	Mobile Wallet	2026-05-05 11:00:00.000
6 Mya Mya	09790000005	20000.00	Bank App	2026-05-06 18:30:00.000
7 Aung Aung	09790000001	10000.00	Mobile Wallet	2026-05-07 08:00:00.000

181 -- 6. RECHARGE HISTORY

182

183

184

185 SELECT

186 c.customer_full_name,

187 s.msisdn,

188 r.recharge_amount,

189 r.recharge_channel,

190 r.recharge_created_at

191 FROM customers c

192 INNER JOIN subscribers s

193 ON c.customer_id = s.customer_id

194 INNER JOIN recharge_transactions r

195 ON s.subscriber_id = r.subscriber_id

196 ORDER BY r.recharge_created_at;

197

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

customer_full_name	msisdn	total_recharge
1 Aung Aung	09790000001	22000.00
2 Mya Mya	09790000005	20000.00
3 Su Su	09790000002	15000.00
4 Hla Hla	09790000004	12000.00
5 Kyaw Kyaw	09790000003	3000.00

customer_full_name	msisdn	total_data_usage_mb
1 Aung Aung	09790000001	3550.75
2 Mya Mya	09790000005	3200.90

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```
198
199  -- 7. TOTAL RECHARGE BY CUSTOMER
200
201
202 SELECT
203     c.customer_full_name,
204     s.msisdn,
205     SUM(r.recharge_amount) AS total_recharge
206 FROM customers c
207 INNER JOIN subscribers s
208 ON c.customer_id = s.customer_id
209 INNER JOIN recharge_transactions r
210 ON s.subscriber_id = r.subscriber_id
211 GROUP BY c.customer_full_name, s.msisdn
212 ORDER BY total_recharge DESC;
213
```

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

	customer_full_name	msisdn	total_data_usage_mb
1	Aung Aung	09790000001	3550.75
2	Mya Mya	09790000005	3200.90
3	Su Su	09790000002	2300.75
4	Hla Hla	09790000004	1750.30
5	Kyaw Kyaw	09790000003	400.00

```
214
215  -- 8. TOTAL DATA USAGE BY CUSTOMER
216
217
218 SELECT
219     c.customer_full_name,
220     s.msisdn,
221     SUM(d.total_data_mb) AS total_data_usage_mb
222 FROM customers c
223 INNER JOIN subscribers s
224 ON c.customer_id = s.customer_id
225 INNER JOIN data_usage d
226 ON s.subscriber_id = d.subscriber_id
227 GROUP BY c.customer_full_name, s.msisdn
228 ORDER BY total_data_usage_mb DESC;
229
```

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

	customer_full_name	msisdn	total_data_usage_mb
1	Aung Aung	09790000001	3550.75
2	Mya Mya	09790000005	3200.90
3	Su Su	09790000002	2300.75
4	Hla Hla	09790000004	1750.30
5	Kyaw Kyaw	09790000003	400.00

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```
230
231 -- 9. HIGH RECHARGE CUSTOMERS
232
233
234 SELECT
235     c.customer_full_name,
236     s.msisdn,
237     SUM(r.recharge_amount) AS total_recharge
238 FROM customers c
239 INNER JOIN subscribers s
240 ON c.customer_id = s.customer_id
241 INNER JOIN recharge_transactions r
242 ON s.subscriber_id = r.subscriber_id
243 GROUP BY c.customer_full_name, s.msisdn
244 HAVING SUM(r.recharge_amount) > 10000;
245
```

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

	customer_full_name	msisdn	total_recharge
1	Aung Aung	09790000001	22000.00
2	Su Su	09790000002	15000.00
3	Hla Hla	09790000004	12000.00
4	Mya Mya	09790000005	20000.00

```
246
247 -- 10. FINAL BUSINESS REPORT
248
249
250 SELECT
251     c.customer_full_name AS Customer_Name,
252     c.city,
253     s.msisdn,
254     s.package_type,
255     s.status,
256
257     ISNULL(
258         (SELECT SUM(r.recharge_amount)
259          FROM recharge_transactions r
260          WHERE r.subscriber_id = s.subscriber_id), 0
261     ) AS total_recharge,
262
263     ISNULL(
264         (SELECT SUM(d.total_data_mb)
265          FROM data_usage d
266          WHERE d.subscriber_id = s.subscriber_id), 0
267     ) AS total_data_usage_mb
268
269 FROM customers c
270 INNER JOIN subscribers s
271 ON c.customer_id = s.customer_id
272 ORDER BY total_recharge DESC;
```

100 % No issues found Ln: 32, Ch: 3 SPC LF UTF-8

	Customer_Name	city	msisdn	package_type	status	total_recharge	total_data_usage_mb
1	Aung Aung	Yangon	09790000001	Prepaid	Active	22000.00	3550.75
2	Mya Mya	Yangon	09790000005	Postpaid	Active	20000.00	3200.90
3	Su Su	Mand...	09790000002	Postpaid	Active	15000.00	2300.75
4	Hla Hla	Bago	09790000004	Prepaid	Active	12000.00	1750.30
5	Kyaw Kyaw	Naypy...	09790000003	Prepaid	Inac...	3000.00	400.00

Query executed successfully. localhost\SQLEXPRESS (15.0 ... LINLAEPHYU\Asus (65) TelecomDataMappingLab 00:00:02 Row: 1, Col: 1 68 rows